



בש - איוול

ANOMALY DETECTION & RCA
SHOPX

06:08		MAD FALSE EARLY ALERT cart http_5xx_rate raw crossing; baseline FP=297/720, failed quality gate
06:30		RELIABLE LOG GC overhead warning, pause=713ms heap=93%
06:33		CART LOG ProductCatalogCache eviction failed: heap pressure too high
14:40		CART METRIC http_p99_latency_ms crosses robust threshold with low baseline FP
16:26		CART METRIC memory_usage_bytes crosses robust threshold
17:50		CART METRIC jvm_gc_pause_ms_avg crosses threshold
19:59		CART LOG OOMKilled: memory limit exceeded
20:41		SUSTAINED 5XX IMPACT corrected 5xx detector fires after restart/downstream symptoms

01

WHEN - 6:30 / 14:40 UTC

Log xuất hiện trước: **GC warning** lúc **06:30:32** với **heap 93%**, **cache eviction failure** lúc **06:33:57**. Metric Anomaly đáng tin đầu tiên là **cart p99 latency** lúc **14:40**.

02

WHERE - cart-service

Template **GC/cache/OOM** xuất hiện trước lỗi downstream.

03

HOW - heap/cache pressure

root-cause dự đoán là **cart-service** bị **heap/memory pressure**, có thể liên quan **ProductCatalogCache retention/eviction failure**. Metric **memory_usage_bytes** không tự chứng minh vượt limit vì max khoảng 1.70GB trong khi limit là 2.15GB. Bằng chứng vượt limit đến từ log **Container OOMKilled: memory limit exceeded**.

EDA

Các service đã load: ['cart', 'order', 'payment', 'gateway', 'product']

- cart: 2820 dòng, từ 2026-06-01 00:00:00+00:00 đến 2026-06-01 23:59:30+00:00
- order: 2820 dòng, từ 2026-06-01 00:00:00+00:00 đến 2026-06-01 23:59:30+00:00
- payment: 2820 dòng, từ 2026-06-01 00:00:00+00:00 đến 2026-06-01 23:59:30+00:00
- gateway: 2820 dòng, từ 2026-06-01 00:00:00+00:00 đến 2026-06-01 23:59:30+00:00
- product: 2820 dòng, từ 2026-06-01 00:00:00+00:00 đến 2026-06-01 23:59:30+00:00

Kích thước bảng sau khi merge: (2820, 27)

ĐỘ PHỦ METRIC

2,820

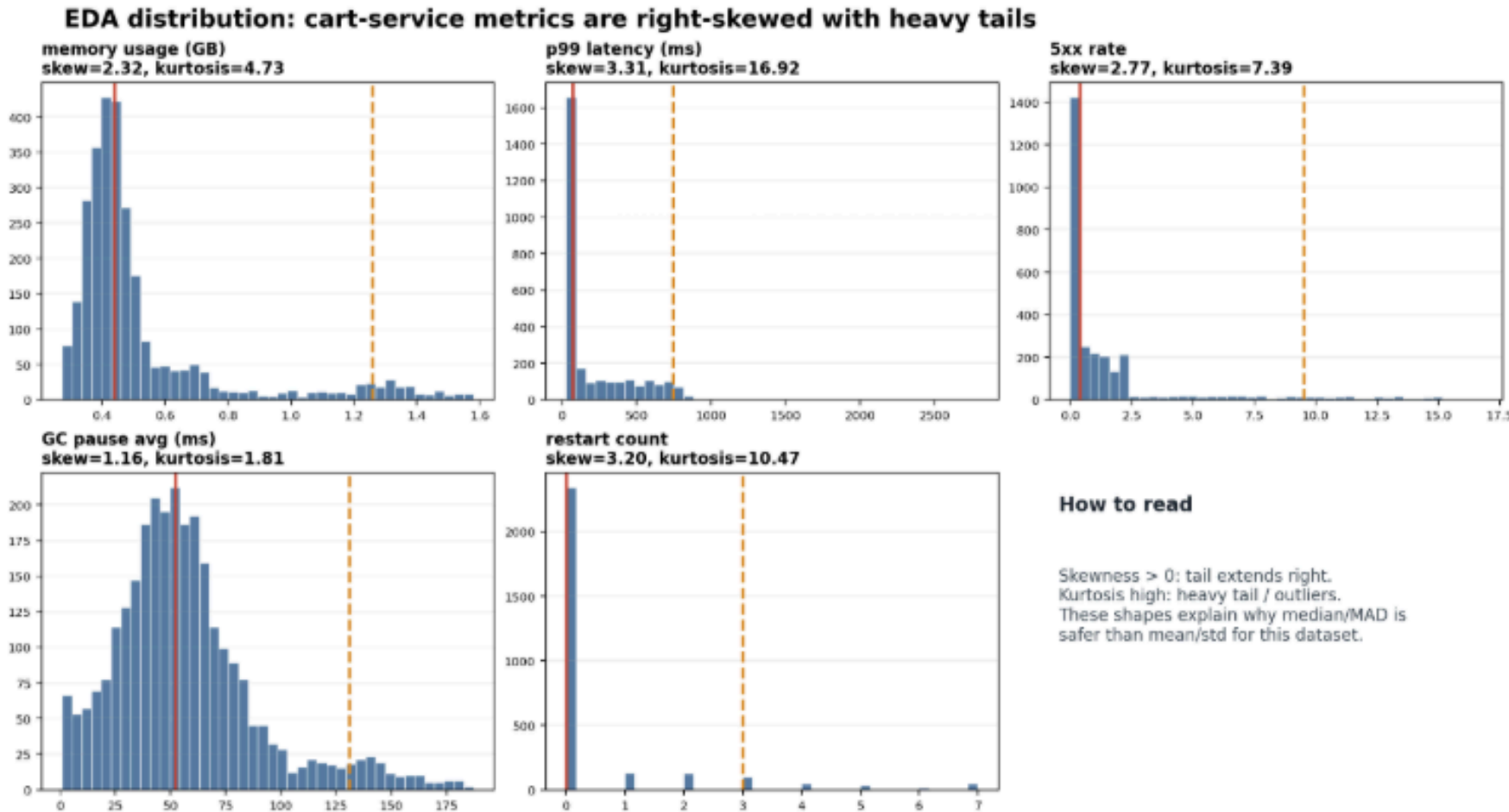
dòng / service, interval 30 giây, đủ một ngày

GIÁ TRỊ THIỂU

0%

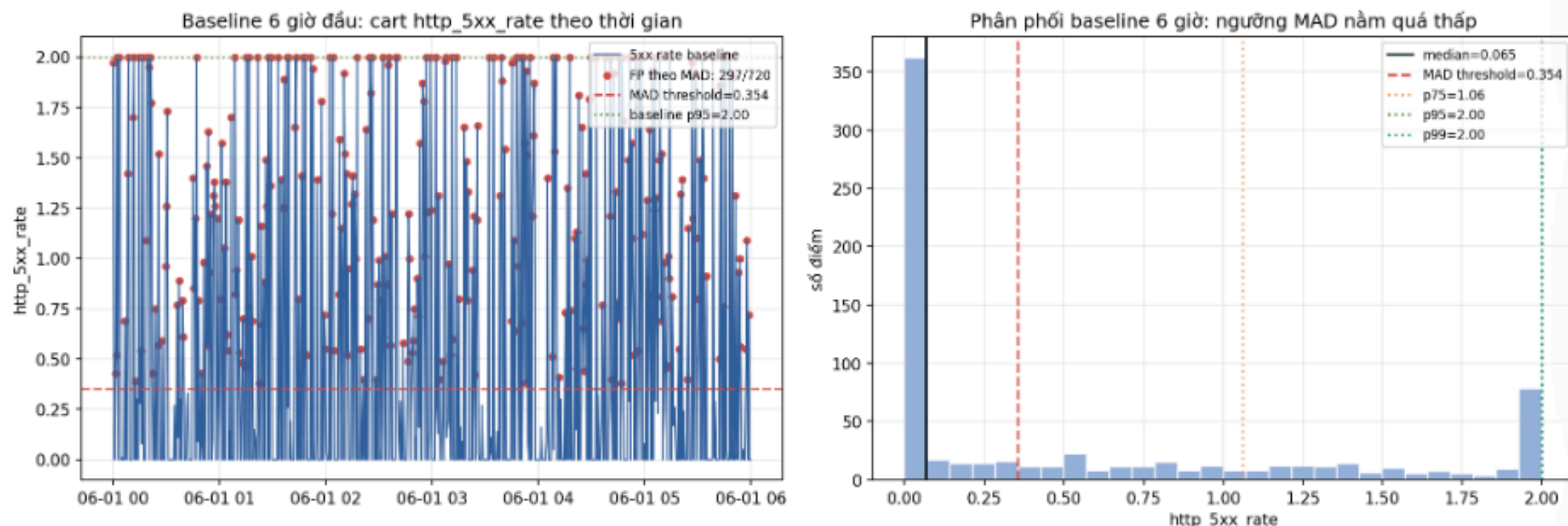
bảng missing trong notebook: các metric chính đều có missing_count = 0

METRIC CART	ĐỘ LỆCH	KURTOSIS	P50	P99
memory_usage_bytes	2.32	4.73	471329720.00	1581406523.33
http_p99_latency_ms	3.31	16.92	74.00	1699.88
http_5xx_rate	2.77	7.39	0.38	14.43
jvm_gc_pause_ms_avg	1.16	1.81	52.10	162.38
container_restart_count	3.20	10.47	0.00	7.00



Baseline 6 giờ là cửa sổ hiệu chỉnh detector, nhưng cũng là nơi lộ điểm yếu của MAD trên cart 5xx.

Audit baseline 6 giờ cho cart-service/http_5xx_rate



VÌ SAO DỪNG 6 GIỜ?

Dùng 6 giờ đầu làm baseline vì đây là cửa sổ trước giai đoạn OOM/restart, đủ 720 điểm ở interval 30 giây để tính median, MAD và percentile.

ĐIỂM YẾU THẤY NGAY

Median baseline chỉ 0.065, nhưng p75=1.06 và p95/p99=2.00. Ngưỡng MAD 0.354 nằm quá thấp so với nhiều baseline.

KẾT LUẬN AUDIT

MAD flag 297/720 điểm trong chính baseline. Vì vậy 06:08 chỉ là lần vượt ngưỡng thô, không phải mốc RCA đáng tin.

ANOMALY DETECTORS & WHY

01 Robust MAD - Primary

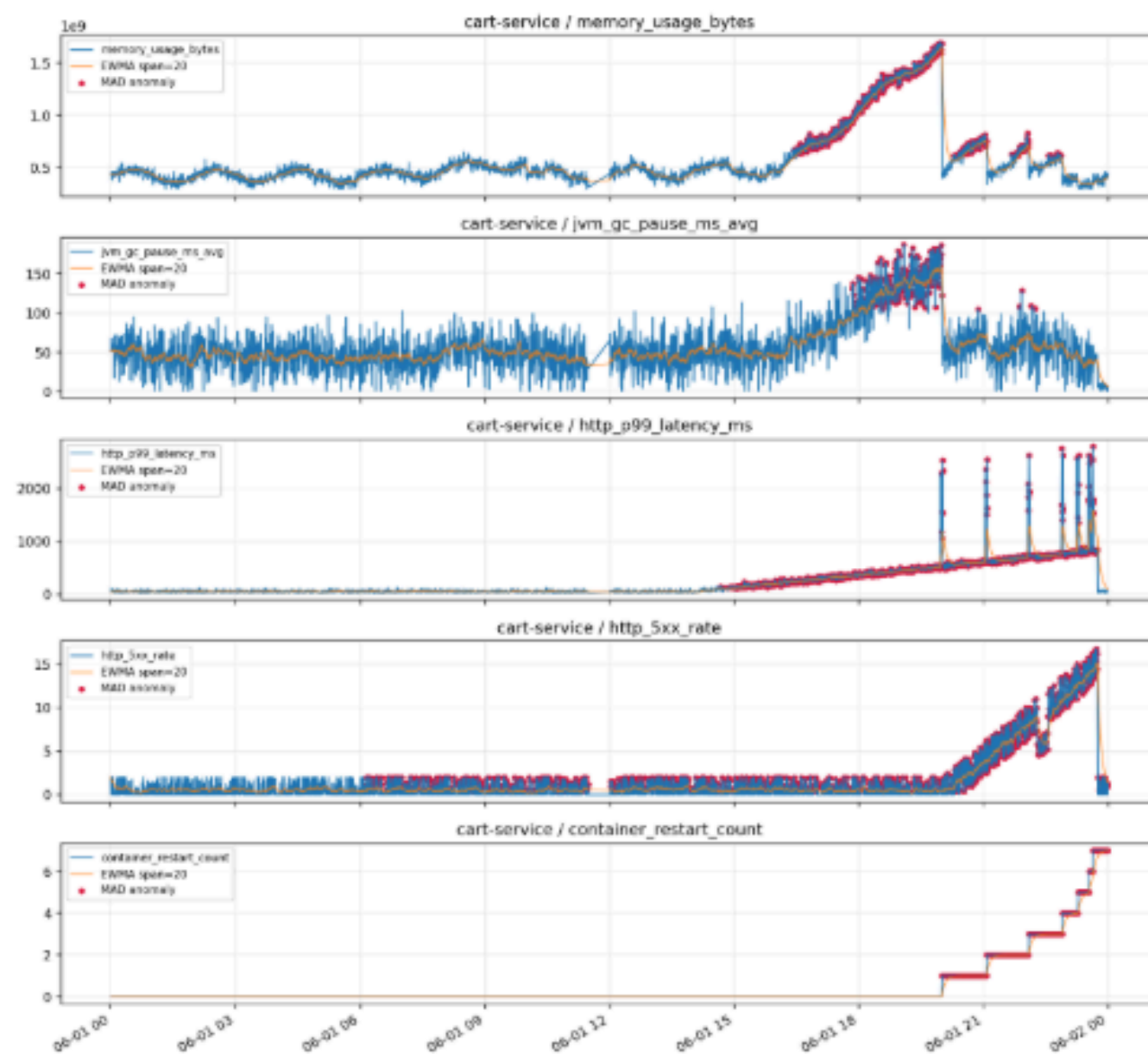
- Các metrics có phân phối right-skewed và heavy-tail; MAD đảm bảo tính robust và explainable cho từng metric.
- Sử dụng 6 giờ đầu tiên làm baseline; $\text{threshold} = \text{median} + 3 * 1.4826 * \text{MAD}$. Đây là primary detector dành cho p99 latency, memory, GC pause, và restart.

02 Isolation Forest

- Detects joint abnormal state khi nhiều metrics cùng dịch chuyển (shift) với nhau.
- Train trên 6 giờ đầu tiên cho mỗi service; score toàn bộ timeline; sử dụng làm confirmation, không dùng làm RCA anchor (vì khó giải thích rõ ràng hơn cho từng metric).

03 EWMA

- Tốt cho việc smoothing noise và hiển thị drift, nhưng quá noisy để dùng làm decision rule.
- Chạy EWMA(span=20) trên các metric series; so sánh với robust threshold; chỉ sử dụng cho mục đích trend visualization.



ÁP LỰC BỘ NHỚ

16:26

memory_usage_bytes vượt ngưỡng MAD

ÁP LỰC GC

17:50

GC pause 131.8ms > ngưỡng 104.3ms

LỖI NHÌN THẤY ĐƯỢC

20:00

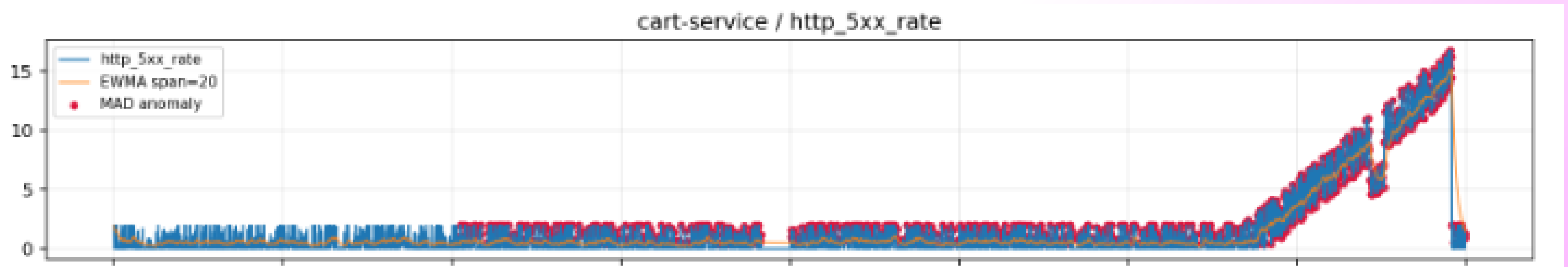
restart_count từ 0 lên 1 sau OOMKilled

Exact robust MAD evidence for cart-service anomaly ordering

UTC time	metric	value	baseline median	MAD threshold	score
06:08:00	http_5xx_rate	1.03	0.07	0.35	10.0?
14:40:00	http_p99_latency_ms	148.70	50.10	122.82	4.1?
16:26:00	memory_usage_bytes	0.62 GB	0.39 GB	0.57 GB	3.8635425810:
17:50:30	jvm_gc_pause_ms_avg	131.80	45.40	104.33	4.4?
20:00:00	container_restart_count	1	0	0	n/a

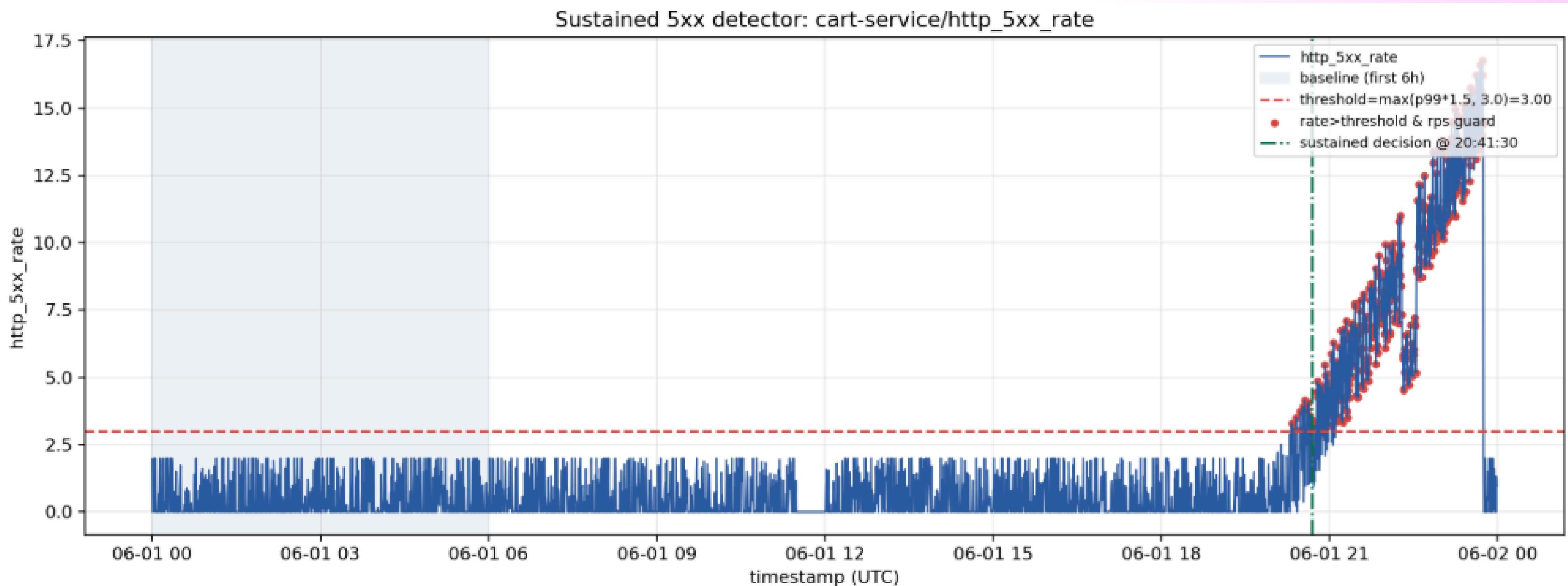
AUDITING - MAD FAILS FOR 5XX

http_5xx_rate crosses threshold at 06:08 → Raw crossing, not evidentiary for anomaly detection



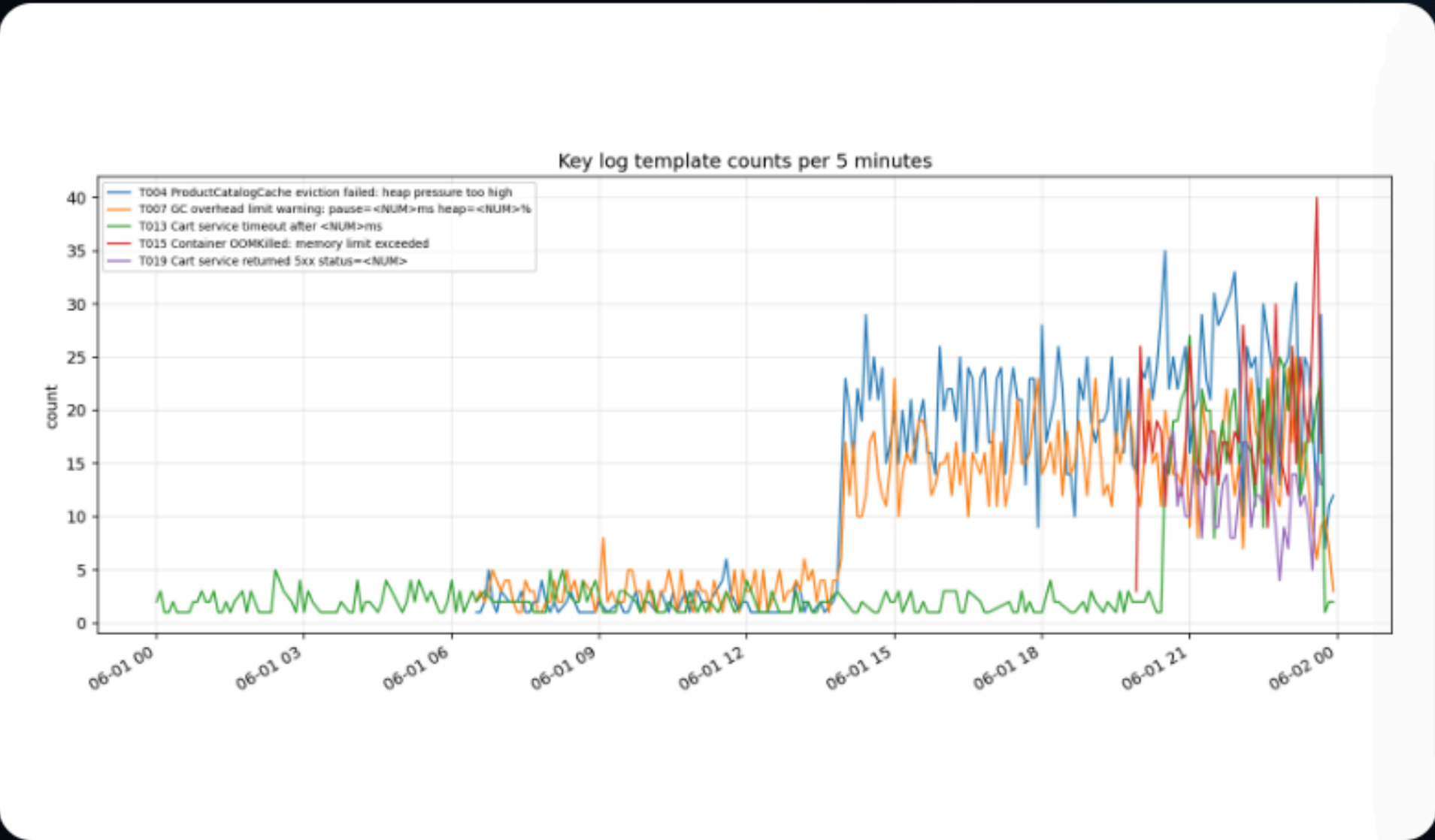
Baseline false positives quá cao đối với metric này -> đã chuyển sang sử dụng sustained detector thay thế:

1. Rule (not ML): threshold = $p99 \times 1.5 = 3.00$.
2. Persistence: 5 trong số 10 points gần nhất vượt quá threshold (5 phút).
3. Volume guard: request rate $\geq$ baseline p50.
4. Result: baseline FP = 0; detect tại thời điểm 20:41; impact_signal = True, supports_rca_chain = False.



5xx sustained là bằng chứng tác động người dùng, không phải móc bắt đầu nguyên nhân gốc.

Log có cấu trúc và nhất quán; Drain3 dùng để template hóa tham số động và đếm pattern theo thời gian.



ID	SVC	LEVEL	SỐ LẦN	THẤY ĐẦU TIÊN	TEMPLATE
T004	cart-service	WARN	2671	2026-06-01T06:33:57.795000+00:00	ProductCatalogCache eviction failed: heap pressure too high
T007	cart-service	WARN	2084	2026-06-01T06:30:32.992000+00:00	GC overhead limit warning: pause=<NUM>ms heap=<NUM>%
T013	order-service	WARN	1080	2026-06-01T00:00:14.987000+00:00	Cart service timeout after <NUM>ms
T015	cart-service	FATAL	823	2026-06-01T19:59:31.047000+00:00	Container OOMKilled: memory limit exceeded
T019	order-service	ERROR	466	2026-06-01T00:00:14.987000+00:00	Cart service returned 5xx status=<NUM>

Drain3 logs khá gọn. Drain3 không “cứu data bản”; nó biến message có `userId/status/duration/heap/pause` thành template đếm được.

RCA DRILL DOWN

Hypothesis

- Cart heap/cache pressure precedes OOM and restart.
- GC warning + cache eviction failure are earliest service-specific signals.
- Downstream timeouts/5xx appear after restarts.

Root cause evidence:

```
4310 -service, "pod": "cart-service-7d9f8b-xaji0", "trace_id": "4243a202902cee32", "message": "Checkout completed orderId=ORD-15191728", "duration_ms": 93}
4311 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "bd80e4260ba8287f", "message": "DB query executed table=cart rows=282", "duration_ms": 22}
4312 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "b25f56d0cee8fafc", "message": "Connection pool nearing limit pool=db connections=47/50"}
4313 -service, "pod": "cart-service-7d9f8b-thv3a", "trace_id": "b8cd1366b399983c", "message": "Connection pool nearing limit pool=db connections=48/50"}
4314 -service, "pod": "cart-service-7d9f8b-xaji0", "trace_id": "8ac269e6d4e5cb53", "message": "DB query executed table=cart rows=37", "duration_ms": 27}
4315 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "6054e1e1d97b3f0c", "message": "Connection pool nearing limit pool=db connections=47/50"}
4316 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "0cdea5b5c5fffaad", "message": "Checkout completed orderId=ORD-95085610", "duration_ms": 170}
4317 -service, "pod": "cart-service-7d9f8b-hsahx", "trace_id": "096efdeaf8c76fec", "message": "DB query executed table=cart rows=117", "duration_ms": 15}
4318 -service, "pod": "cart-service-7d9f8b-hsahx", "trace_id": "2e04addd79e3cbdd", "message": "Checkout completed orderId=ORD-31010797", "duration_ms": 103}
4319 -service, "pod": "cart-service-7d9f8b-hsahx", "trace_id": "dbcb5ad5d9ff1a04", "message": "Health check response time elevated endpoint=/health latency=207ms"}
4320 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "3255a3dac081fff4", "message": "ProductCatalogCache eviction failed: heap pressure too high", "cache_size_mb": 1457.6}
4321 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "c3ffdbebaef35fe3", "message": "Health check response time elevated endpoint=/health latency=226ms"}
4322 -service, "pod": "cart-service-7d9f8b-thv3a", "trace_id": "cab40e85664c9df4", "message": "Item added to cart for userId=30028", "duration_ms": 11}
4323 -service, "pod": "cart-service-7d9f8b-thv3a", "trace_id": "39d3add114153ea1", "message": "Checkout completed orderId=ORD-65751198", "duration_ms": 157}
4324 -service, "pod": "cart-service-7d9f8b-xaji0", "trace_id": "0b8ed5dd33a3e5bc", "message": "Checkout completed orderId=ORD-14845702", "duration_ms": 98}
4325 -service, "pod": "cart-service-7d9f8b-xaji0", "trace_id": "3fdce8013011ead6", "message": "Item added to cart for userId=25842", "duration_ms": 17}
4326 -service, "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "7e5bef017b4fdce8", "message": "Item added to cart for userId=33324", "duration_ms": 10}
4327 -service, "pod": "cart-service-7d9f8b-xaji0", "trace_id": "edf24aae8b44383b", "message": "Health check passed", "duration_ms": 1}
4328 -service, "pod": "cart-service-7d9f8b-xaji0", "trace_id": "cb05c4d6fa4cc2be", "message": "Item added to cart for userId=51256", "duration_ms": 7}
4329 -service, "pod": "cart-service-7d9f8b-hsahx", "trace_id": "0c34a09fc9ed6e1e", "message": "Cache warm-up slower than expected startup_ms=2591"}
```


RCA DRILL DOWN

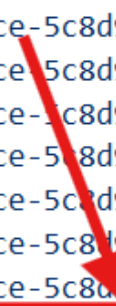
Impact evidence:

g1 > logs > {} cart-service.log.jsonl

```
17773 01T19:58:36.759Z", "level": "WARN", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "a9e572e0c516f207", "message": "GC overhead limit warning: pause=500ms"
17774 01T19:58:36.777Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "ac5eadfcd6a6bd3a", "message": "Checkout completed orderId=ORD-21258500"
17775 01T19:58:36.838Z", "level": "WARN", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "2567d1daaae8c9db", "message": "Health check response time elevated end"
17776 01T19:58:39.145Z", "level": "WARN", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "709c2bdf46ab6fca", "message": "ProductCatalogCache eviction failed: he"
17777 01T19:58:41.335Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "ef7edf1130fed1c5", "message": "DB query executed table=cart rows=192",
17778 01T19:58:42.757Z", "level": "WARN", "service": "cart-service", "pod": "cart-service-7d9f8b-hsahx", "trace_id": "c71002babde9d61f", "message": "Slow response detected endpoint=/api/ca"
17779 01T19:58:45.444Z", "level": "WARN", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "5243d32dc2ac986a", "message": "Health check response time elevated end"
17780 01T19:58:47.318Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "9ae7ee3d619ced8a", "message": "Checkout completed orderId=ORD-86251139"
17781 01T19:58:49.042Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "8b3c1d8d7b0b217c", "message": "Health check passed", "duration_ms": 2}
17782 01T19:58:53.670Z", "level": "WARN", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "205b8ade724c4ccc", "message": "GC overhead limit warning: pause=511ms"
17783 01T19:58:56.477Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "bdb9db812d85f9ca", "message": "Item added to cart for userId=53765", "c"
17784 01T19:59:13.548Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-xaji0", "trace_id": "2dfcaa397ea03390", "message": "Application starting up version=2.4.1"
17785 01T19:59:15.939Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "2cf2debb6f406a81", "message": "OutOfMemoryError imminent: available h"
17786 01T19:59:16.133Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-xaji0", "trace_id": "e9fcbfe86b58da0f", "message": "Application starting up version=2.4.1",
17787 01T19:59:18.045Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "cfcad2eac4bfbf85", "message": "Application starting up version=2.4.1",
17788 01T19:59:24.821Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "884f03a8d6e168e8", "message": "OutOfMemoryError imminent: available h"
17789 01T19:59:26.256Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-hsahx", "trace_id": "d2dbaaaadb6cc0eb", "message": "Upstream connection refused host=produ"
17790 01T19:59:27.064Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "ab5f9a1153d766e4", "message": "Application starting up version=2.4.1",
17791 01T19:59:31.047Z", "level": "FATAL", "service": "cart-service", "pod": "cart-service-7d9f8b-hsahx", "trace_id": "3afddc558cb7b87c", "message": "Container OOMKilled: memory limit exce"
17792 01T19:59:34.076Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-hsahx", "trace_id": "b4011d027084e258", "message": "Application starting up version=2.4.1",
17793 01T19:59:34.528Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "aeb4192b9fbbebb89", "message": "OutOfMemoryError imminent: available h"
17794 01T19:59:39.353Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "e95e82a29f9bd096", "message": "Application starting up version=2.4.1",
17795 01T19:59:39.795Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "5ad84cd6bd7cd51d", "message": "Upstream connection refused host=produ"
17796 01T19:59:42.383Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "c9c830e7a2faac56", "message": "Application starting up version=2.4.1",
17797 01T19:59:43.824Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "6ef3657d9ccfcbfe", "message": "Application starting up version=2.4.1",
17798 01T19:59:44.089Z", "level": "FATAL", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "bfc632e0b3cdb705", "message": "Container OOMKilled: memory limit exce"
17799 01T19:59:47.891Z", "level": "FATAL", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "bda00fcecc1f1eb8", "message": "Container OOMKilled: memory limit exce"
17800 01T19:59:49.555Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "dd2e43f4a92b5793", "message": "Upstream connection refused host=produ"
17801 01T19:59:53.142Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "cedade6e54fad5ee", "message": "OutOfMemoryError imminent: available h"
17802 01T19:59:57.289Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "ab2f6fbba1beaaa9", "message": "Application starting up version=2.4.1",
17803 01T19:59:58.591Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "dc95f9e47c319ee0", "message": "OutOfMemoryError imminent: available h"
17804 01T20:00:01.616Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-thv3a", "trace_id": "e951880a0d0a9f10", "message": "OutOfMemoryError imminent: available h"
17805 01T20:00:04.247Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-xaji0", "trace_id": "e3e5ea2e40a2a8ae", "message": "OutOfMemoryError imminent: available h"
17806 01T20:00:07.986Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-xaji0", "trace_id": "379cfecd56bf3527", "message": "Upstream connection refused host=produ"
17807 01T20:00:13.246Z", "level": "ERROR", "service": "cart-service", "pod": "cart-service-7d9f8b-y6dpb", "trace_id": "6dccb3bc25db7f86", "message": "OutOfMemoryError imminent: available h"
17808 01T20:00:16.021Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-hsahx", "trace_id": "32db2cfad4a8eded", "message": "Application starting up version=2.4.1",
17809 01T20:00:17.483Z", "level": "INFO", "service": "cart-service", "pod": "cart-service-7d9f8b-xaji0", "trace_id": "67e3cd89dff64cb3", "message": "Application starting up version=2.4.1",
```


RCA DRILL DOWN

Impact evidence:



```
6158 T20:28:22.919Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "ebf0aabafc5bcd37", "message": "Cart service call succeeded", "duration_
6159 T20:28:36.143Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "bcf627d5df647155", "message": "Order created orderId=ORD-57949753 userI
6160 T20:28:42.735Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "43c47fe49650adcd", "message": "Health check passed", "duration_ms": 3}
6161 T20:28:55.676Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "4ec2cfddda2bc15a", "message": "Cart service call succeeded", "duration_
6162 T20:28:58.994Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-6989s", "trace_id": "8abe13bfec1d2cae", "message": "Health check passed", "duration_ms": 2}
6163 T20:29:15.900Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "8ebac8182c15f7fd", "message": "Cart service call succeeded", "duration_
6164 T20:29:39.348Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "a92150d61409dbf9", "message": "Order created orderId=ORD-63175318 userI
6165 T20:30:00.520Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "b8f900e69ae79afe", "message": "Cart service timeout after 2336ms"}
6166 T20:30:03.634Z", "level": "ERROR", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "8c54f448ed8c1fcb", "message": "Cart service returned 5xx status=500"}
6167 T20:30:04.304Z", "level": "ERROR", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "4ba011f1d7734ffc", "message": "Cart service returned 5xx status=500"}
6168 T20:30:22.660Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "16a0ad15ac1954d0", "message": "Health check passed", "duration_ms": 1}
6169 T20:30:24.775Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "bf4f18367febb9da", "message": "Cart service timeout after 939ms"}
6170 T20:30:31.651Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "efe2ebfefe2ee6c4", "message": "Health check passed", "duration_ms": 3}
6171 T20:30:33.463Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "eaef4ba494b4093d", "message": "Cart service timeout after 1912ms"}
6172 T20:30:34.771Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "bcbefaab40e3aa0b", "message": "Cart service timeout after 633ms"}
6173 T20:30:37.465Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "6aa3d1c3edcdd94b", "message": "Cart service timeout after 2363ms"}
6174 T20:30:39.293Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "4bb9d54fc68ad2a4", "message": "Cart service timeout after 1457ms"}
6175 T20:30:44.972Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "3ecfdea53baf9bf3", "message": "Health check passed", "duration_ms": 3}
6176 T20:30:54.239Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "7a07868adc49974d", "message": "Cart service timeout after 2672ms"}
6177 T20:31:09.759Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-6989s", "trace_id": "22d2ca5201ef4cde", "message": "Cart service call succeeded", "duration_
6178 T20:31:11.127Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-6989s", "trace_id": "ae1dd3dc29c86c5c", "message": "Cart service call succeeded", "duration_
6179 T20:31:28.078Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "dff15ac47ed5bae6", "message": "Order created orderId=ORD-13303035 userI
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6181 T20:31:35.223Z", "level": "WARN", "service": "order-service", "pod": "order-service-5c8d9a-zqgxq", "trace_id": "b9f145e44fc47b1e", "message": "Cart service timeout after 2979ms"}
6182 T20:32:04.068Z", "level": "INFO", "service": "order-service", "pod": "order-service-5c8d9a-aaxn4", "trace_id": "dcae17cd365a8cb5", "message": "Health check passed", "duration_ms": 1}
6183 T20:32:11.461Z", "level": "ERROR", "service": "order-service", "pod": "order-service-5c8d9a-6989s", "trace_id": "91141ba8c5cd9ebc", "message": "Cart service returned 5xx status=500"}

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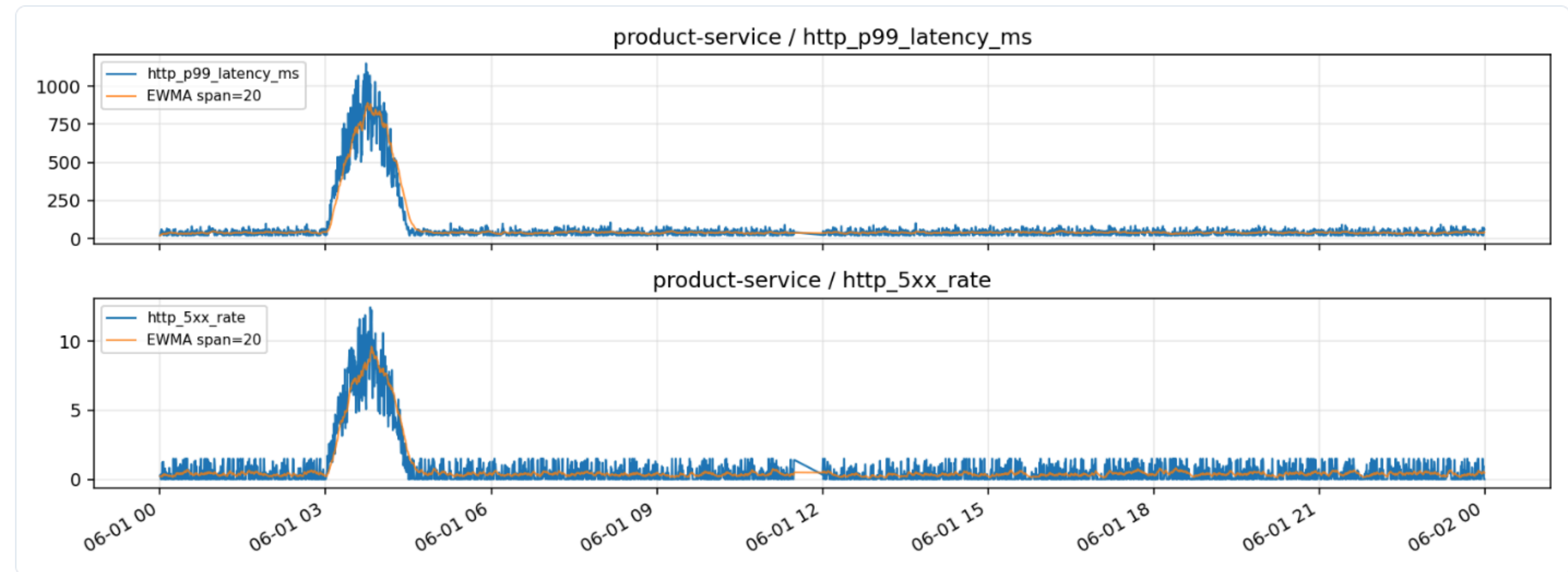

RCA DRILL DOWN

Exclusions

- 06:08 5xx MAD crossing bị loại trừ (baseline FP quá cao).
- EWMA early flags bị loại trừ (chỉ để xem trend).

Product-service không phải RCA chính

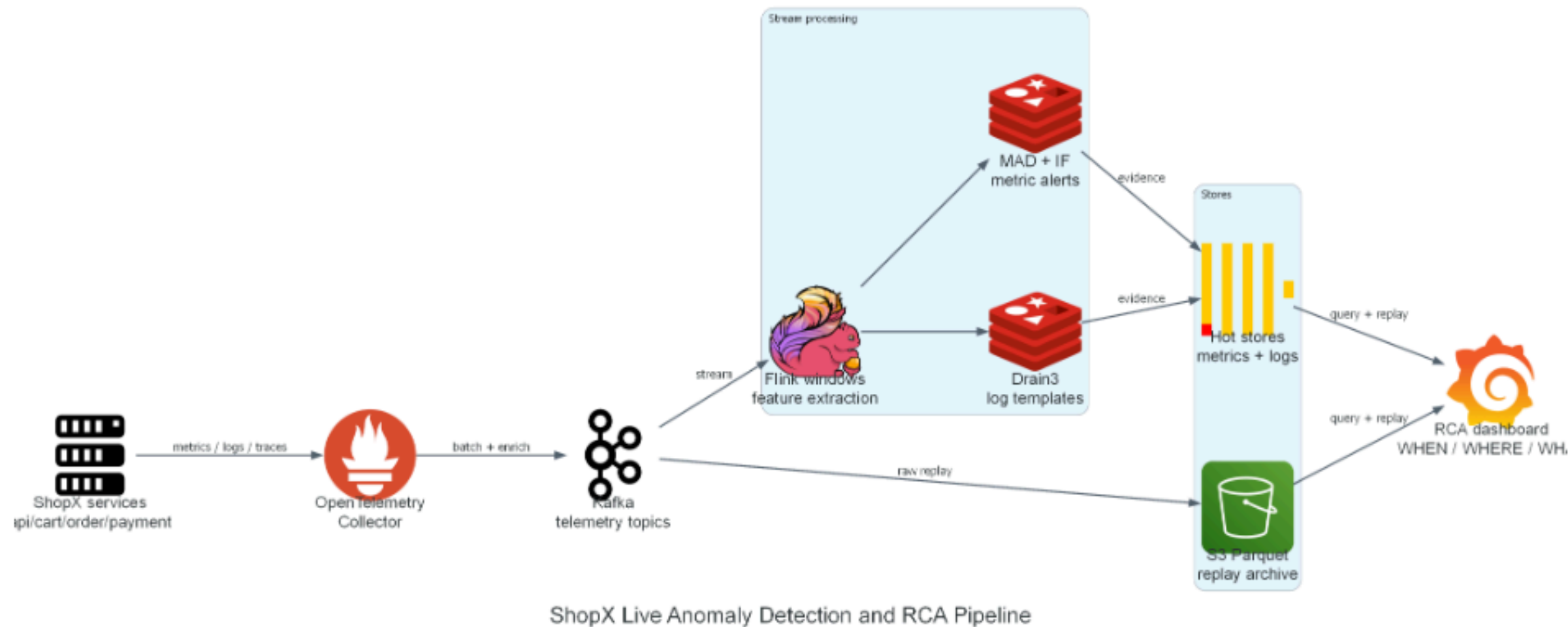
Product-service có dao động và có log connection refused liên quan catalog, nhưng bằng chứng không tạo thành chuỗi restart loop rõ như cart-service. Vì vậy nó được xem là yếu tố phụ hoặc nhiễu cho đến khi có thêm telemetry chứng minh ngược lại.



charts/product-service-metrics.png

Product-service có dao động, nhưng không tạo được chuỗi memory -> OOMKilled -> restart rõ như cart-service.

REAL WORLD DATA PIPELINE



INGEST

OTel -> Kafka

Collector enriches; Kafka buffers and enables replay.

DETECT

Flink + MAD/IF

Windowed features, metric alerts, and Drain3 templates.

RCA

Dashboard

Hot stores and replay archive support evidence ordering.

REAL WORLD DATA PIPELINE

Production flow: telemetry được ingest liên tục, detector chạy theo window, RCA ghép evidence theo thời gian.

1

Emit

Services emit metrics, logs, traces with service/pod labels.

2

Ingest

OpenTelemetry Collector batches, enriches, filters, and routes signals.

3

Buffer

Kafka topics decouple services from processing and enable replay.

4

Detect

Flink computes windows; MAD + IF score metrics; Drain3 templates logs.

5

RCA

Evidence timeline ranks origin candidate and downstream blast radius.

Early alert target: memory slope + GC pause + cache eviction template count + cart p99/5xx.



בש - איוול

ANOMALY DETECTION & RCA
SHOPX